Shipgen Platform

Product Vision vs Codebase Implementation Gap Analysis

AI-Powered Smart Logistics Operations Platform

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Source: Finalized product deck + Fleetbase codebase audit

This document compares the finalized product vision (10-slide deck) against the current Shipgen codebase to identify completed capabilities, partial implementations, and gaps requiring future development.

1. Executive Summary

Shipgen is positioned in the product deck as a unified, AI-native logistics platform with three intelligent engines — **FleetOps**, **Yard OS**, and **Tollgate OS** — powered by a centralized AI Engine. The current codebase delivers a **substantial operational foundation** built on Fleetbase with embedded Yard (YMS) and Parking (PMS) modules, web console, and mobile apps.

Key finding: Core operational workflows across fleet, yard, and parking are largely implemented. The primary gap is between **marketing AI capabilities** (predictive ML, NLP copilot, fraud AI, auto-dispatch) and **actual implementation** (rule-based heuristics, VROOM/OSRM optimization, EasyOCR plate recognition, manual gate workflows).

Platform Area	Vision Scope	Implementation	Completion
Unified Platform (Single Login)	One login, role-based access across modules	React console + IAM + engine switcher	Complete
FleetOps Engine	AI route, driver, maintenance, GPS, POD	Full FleetOps + orchestrator; heuristic (not ML) AI	Partial
Yard OS Engine	AI queue, dock, labor, OCR gate, heatmap	Full 15-status lifecycle; rule-based recommendations	Partial
Tollgate OS Engine	ANPR, fraud AI, barrier automation, digital toll	Parking PMS: ticketing, OCR upload, manual ops	Partial
Central AI Engine	Predictive analytics, NLP Q&A;, cross-module insights	Not implemented; rule-based alerts only	Not Started
End-to-End Lifecycle	14-step digitized supply chain	Steps 1–13 largely covered; cross-engine automation partial	Partial
Mobile + Web	Role-based field and floor apps	Expo driver app + YMS mobile roles	Complete
Enterprise / API	API-first, on-prem, audit, compliance	REST APIs, webhooks, audit trails, Docker on-prem	Complete
Supporting Engines	Storefront, Ledger, Pallet, Registry	Full backend + console UI present	Complete

Overall estimated completion vs product vision: ~72% operational foundation, ~35% AI-native vision as marketed in the deck.

2. Analysis Methodology

- **Product source:** Finalized Canva deck — _AI-Powered-Smart-Logistics-Operations-Platform_.pdf (10 slides, July 2026).
- **Technical source:** Fleetbase monorepo audit — Laravel packages, YMS FastAPI, Parking FastAPI, React console, Expo mobile.
- **Reference BRD:** docs/SHIPGEN-BRD.md v2.0 (June 2026) — authoritative business requirements.
- **Status definitions:** Complete = production-ready end-to-end; Partial = core exists but vision gaps remain; Not Started = no meaningful implementation; Out of Scope = explicitly deferred per BRD.

3. Platform Architecture

The product deck describes a three-layer architecture: Single Login → Operational Modules (FleetOps, Yard OS, Tollgate OS) → AI & Insights. The codebase maps as follows:

Vision Layer	Codebase Mapping	Status
Single Login + RBAC	core-api auth, IAM, 2FA, org switching, engine switcher	**Complete**
FleetOps	packages/fleetops + frontend /fleet-ops/* + mobile driver module	**Complete**
Yard OS	yms/ FastAPI + frontend /yard/* + mobile (yard)/*	**Complete**
Tollgate OS	Parking management/ → embedded /parking/* (PMS)	**Partial**
AI & Insights Layer	Rule-based YMS alerts; no central AI engine or NLP	**Not Started**
REST APIs + IoT readiness	1,350+ endpoints, telematics webhooks, device APIs	**Complete**
Cloud / On-prem deploy	Docker, OSRM, gateway, ON-PREM docs	**Complete**

4. Module 01 — FleetOps (Intelligent Fleet Management)

Product vision: AI route optimization, AI driver recommendation, predictive maintenance, live GPS & digital POD. Workflow: Plan → Track → Predict → Confirm.

Feature (Product Deck)	Implementation Evidence	Status	Gap Notes
Order management & dispatch	OrderController, dispatch/bulk-dispatch, order board Kanban	**Complete**	Full lifecycle with workflow engine
AI Route Optimization	OSRM.php routing + VroomOrchestrationEngine VRP solver	**Partial**	Algorithmic VRP, not ML; no live traffic AI
AI Driver Recommendation	DriverAssignmentEngine (greedy best-fit: skills, geo, shift)	**Partial**	Heuristic matching, not ML performance model
Predictive Maintenance	Maintenance schedules, telematics (Samsara/Geotab/Flespi)	**Partial**	Preventive scheduling; no failure prediction ML
Live GPS Tracking	Driver track API, live coordinates, SocketCluster realtime	**Complete**	Console live map + mobile tracking tab
Digital POD	capture-signature/photo/qr endpoints + mobile podService	**Complete**	Signature, photo, QR proof capture
Route planning & orchestrator	Orchestrator workbench, manifests, route sequencing	**Complete**	Greedy + VROOM engines registered
Driver mobile app	frontend_mobile: orders, workflow, POD, offline queue	**Complete**	50+ screens, E2E tested
Fleet / vehicle / maintenance mgmt	Drivers, vehicles, fleets, work orders, parts CRUD	**Complete**	145+ FleetOps console pages
Geofencing & telematics	Geofence events, dwell reports, provider webhooks	**Complete**	Samsara, Geotab, Flespi integrations
Analytics & reports	SQL report builder, fleet-ops/metrics, custom dashboards	**Complete**	Export CSV/XLSX supported
Auto-dispatch (40% faster claim)	Manual dispatch + orchestrator commit; no AI auto-dispatch	**Partial**	Deck KPI not yet measurable

5. Module 02 — Yard OS (Smarter Yard Operations)

Product vision: AI queue optimization, AI dock allocation, AI labor recommendation, OCR gate entry, AI yard heatmap. Workflow: Arrival → Queue → Dock → Loading → Exit.

Feature (Product Deck)	Implementation Evidence	Status	Gap Notes
15-status vehicle lifecycle	yms.py workflow transitions, gate/queue/dock APIs	Complete	DRAFT through EXITED per BRD
Appointments & scheduling	/yard/appointments, calendar, slot booking	Complete	Priority scoring implemented
Gate entry/exit checklists	gate.py: approve/reject, exit verify, audit trail	Complete	Digital checklists replace paper
Virtual queue & priority	queue.py: score = priority×10 + lateness	Complete	Supervisor override with audit
AI Queue Optimization	Deterministic priority formula + rule alerts	Partial	Not ML-based reordering
AI Dock Allocation	Manual/supervisor dock assign + type constraints	Partial	No automated AI dock matcher
AI Labor Recommendation	Labor roster, assign/release, readiness checks	Partial	Manual assignment; no AI staffing model
OCR Gate Entry	Manual plate search + QR; no yard ANPR camera	Not Started	OCR exists in Parking module only
AI Yard Heatmap	Docks schedule heatmap (appointments by hour)	Partial	Schedule heatmap, not congestion AI heatmap
Weighing (tare/gross/net)	queue tare + dock gross weight endpoints	Complete	Auto net calculation
Detention billing	/yard/detention, status workflow, INR costing	Complete	Auto-derived from wait/load timing
Control Tower & alerts	control_tower.py, 9 alert types, 30s refresh	Complete	Rule-based, not predictive
AI Insights / Recommendations	AiInsights.jsx — explicit rule-based mapping	Partial	Labeled 'not predictive AI' in code
Yard map & zone management	/yard/yard, zone occupancy, manual moves	Complete	Hazmat zone rules enforced
Loading ops & exceptions	Start/pause/resume/complete, exception workflow	Complete	Resource gating before loading
YMS mobile (5 roles)	frontend_mobile/(yard)/* gate, queue, docks, etc.	Complete	Role-based tabs per BRD
Yard reports & KPIs	Delay analysis, executive KPIs, PDF export	Complete	CSV/XLSX/PDF exports

6. Module 03 — Tollgate OS (Frictionless Gate Control)

Product vision positions Tollgate OS as AI number plate recognition, fraud detection, digital payments, barrier automation, and real-time lane dashboards. In the codebase, this maps primarily to the **Parking Management System (PMS)** embedded at /parking/*.

Feature (Product Deck)	Implementation Evidence	Status	Gap Notes
Ticket entry & lifecycle	tickets.py: create, pay, exit; Unpaid→Paid→Exited	Complete	Full parking ticket workflow
AI Number Plate Recognition	EasyOCR via image upload (ocr/service.py)	Partial	Upload-based OCR; no camera ANPR lane
AI Fraud Detection	No fraud detection module found	Not Started	Duplicate/stolen plate AI not built
Digital Payments	Cash, UPI, card payment collection	Complete	Payment audit trail logged
Barrier automation	Hardware config UI (mock/seed data); no barrier control	Not Started	BRD: monitoring only, automation deferred
Real-time dashboard	Admin/supervisor/operator dashboards	Complete	Occupancy, revenue, traffic KPIs
Floor & occupancy mgmt	floors.py capacity buckets 2W/4W/heavy	Complete	Auto increment/decrement on entry/exit
Pricing rules	pricing.py per vehicle category	Complete	Admin-configured rates
QR monitoring	QR scan events visible to supervisor	Complete	Live scan event stream
Role-based access (3 roles)	Admin, supervisor, operator route guards	Complete	SSO from Shipgen console
Reports & admin PDF export	Revenue, traffic, category reports	Complete	Admin PDF export tier
Security & compliance audit	audit.py event logging	Complete	Ticket/payment/pricing changes logged

7. Central AI Engine & Cross-Module Intelligence

Feature (Product Deck)	Status	Notes
Centralized AI Engine across all modules	Not Started	No shared ML/AI service layer
'Ask Your AI Anything' NLP interface	Not Started	No LLM integration in codebase
Predictive analytics & risk detection	Not Started	Explicitly out of scope in BRD §22
Automated recommendations (ML)	Partial	Rule-based alert→action mapping in YMS
Fuel consumption anomaly detection	Not Started	Fuel reports exist; no AI anomaly
Executive decision intelligence	Partial	KPI dashboards on-screen; no AI copilot
Cross-engine data pipeline to AI	Partial	Separate DBs per engine; no unified AI lake

8. End-to-End Workflow (14 Steps)

Step	Workflow Stage	Engine	Status
1	Customer Order	Storefront / FleetOps	Complete
2	Fleet Planning	FleetOps orchestrator	Complete
3	Driver Assignment	FleetOps dispatch	Complete
4	Vehicle Tracking	FleetOps + mobile GPS	Complete
5	Yard Appointment	YMS appointments	Complete
6	Gate Entry	YMS gate module	Complete
7	Dock Allocation	YMS docks	Complete
8	Loading	YMS loading ops	Complete
9	Exit	YMS gate exit	Complete
10	Tollgate Verification	Parking PMS	Partial
11	Delivery	FleetOps order complete	Complete
12	Digital POD	FleetOps + mobile	Complete
13	Invoice	Ledger invoices from orders	Complete
14	Analytics	Per-engine reports	Partial

Note: Steps 1–13 operate within individual engines. Automated handoffs between engines (e.g., FleetOps order triggering yard appointment) require manual coordination today.

9. Supporting Platform Modules

Module	Capabilities	Status
Console / IAM	Dashboard, notifications, users, roles, policies, groups, 2FA	**Complete**
Storefront	Catalog, cart, checkout, customer portal, promotions	**Complete**
Ledger	Invoices, wallets, journals, financial reports, payment gateways	**Complete**
Pallet (WMS)	Warehouses, inventory, purchase/sales orders, audits	**Complete**
Developers	API keys, webhooks, sockets, request logs	**Complete**
Registry	Extension marketplace, install/uninstall, developer accounts	**Complete**
Mobile Apps	Driver + 5 yard roles, offline queue, realtime	**Complete**
On-prem / Docker	OSRM, tiles, gateway, installer	**Complete**

10. Business Impact Claims vs Reality

Deck Claim	Current State	Assessment
40% faster dispatch	Manual + orchestrator dispatch	**Partial**
60% less manual work	Digital workflows replace paper in yard/fleet	**Partial**
30% lower fuel cost	Route optimization via OSRM/VROOM	**Partial**
50% reduced waiting time	Queue priority + SLA alerts	**Partial**
99% process visibility	Audit trails + live dashboards	**Complete**

Quantitative KPI claims in the deck are aspirational targets. The platform provides the operational instrumentation to measure them, but baseline benchmarks are not yet documented.

11. Prioritized Gap Roadmap

11.1 High Priority — Align Product Story with Delivery

#	Gap	Recommended Action	Effort
1	Central AI Engine / NLP copilot	Define MVP: rule-based Q&A; over metrics first; LLM phase 2	High
2	Yard OCR gate entry	Reuse Parking EasyOCR service at YMS gate	Medium
3	Cross-engine automation	FleetOps→YMS appointment trigger; detention→Ledger invoice	Medium
4	AI positioning clarity	Rebrand heuristic features or invest in ML models	Low
5	Tollgate fraud detection	Duplicate plate rules + alert workflow	Medium

11.2 Medium Priority — Vision Enhancement

#	Gap	Recommended Action	Effort
6	Predictive maintenance ML	Telematics feature pipeline + anomaly model	High
7	AI auto-dispatch	Orchestrator auto-commit with dispatcher approval gate	Medium
8	Yard congestion heatmap	Real-time zone occupancy visualization layer	Medium
9	ANPR camera integration	Hardware adapter for barrier lanes	High
10	Unified analytics lake	Cross-engine ETL for executive AI insights	High

11.3 Explicitly Out of Scope (per BRD)

- Full ERP replacement
- Carrier self-service booking portal
- Multi-facility enterprise tenancy
- Customer-facing yard tracking portal
- Advanced OR/TMS solver beyond VROOM

12. BRD Acceptance Criteria Status

Criteria Area	Key Test	Status
Platform	Yard-only user cannot open FleetOps	Complete
Platform	Parking-only user cannot open Yard	Complete
Platform	Shipgen admin sees all engines	Complete
FleetOps	Driver sees assigned order on mobile	Complete
FleetOps	Driver cannot skip workflow step	Complete
FleetOps	POD appears on console order detail	Complete
Yard	Appointment→gate→queue without re-entry	Complete
Yard	Call in→dock→labor→loading chain enforced	Complete
Yard	Loading complete→exit holding auto	Complete
Yard	Exit checklist blocks verify	Complete
Yard	Tare/gross/net weights correct	Complete
Yard	Detention on exceeded free time	Complete
Yard	Wait>60min alert fires	Complete
Yard	Mobile gate entry from phone	Complete
Parking	Ticket entry increments occupancy	Complete
Parking	Exit decrements occupancy	Complete
Parking	Revenue report matches payments	Complete
Reporting	CSV export matches filtered list	Complete

BRD acceptance criteria: 18 of 18 checked items have corresponding implementation. Product deck AI features extend beyond BRD scope.

13. Conclusion

The Shipgen codebase is a **mature multi-engine logistics platform** with strong coverage of operational requirements defined in SHIPGEN-BRD.md. FleetOps, Yard OS, and Parking (Tollgate) modules are production-grade with web and mobile clients, comprehensive APIs, audit trails, and role-based access.

The gap between the **product deck vision** and **current implementation** is concentrated in AI-native capabilities: predictive analytics, NLP copilot, fraud AI, automated gate/barrier systems, and cross-module intelligence. These are either not started or implemented as rule-based/heuristic alternatives. Closing this gap is primarily a product positioning and AI engineering investment — not a greenfield platform build.

Appendix A — Key Codebase Paths

Component	Path
Backend API	packages/core-api, packages/fleetops, packages/ledger, packages/storefront, packages/pallet
Yard (YMS)	yms/backend/routers/, yms/frontend_1/src/
Parking (PMS)	Parking management/backend/, Parking management/frontend/
Web Console	frontend/src/ (436+ route screens)
Mobile	frontend_mobile/frontend/ (Expo, 50+ screens)
Business Requirements	docs/SHIPGEN-BRD.md
Technical Specs	documents/LOW-LEVEL-REQUIREMENTS.md, documents/BACKEND-LOW-LEVEL-REQUIREMENTS.md
Product Deck	ppt/_AI-Powered-Smart-Logistics-Operations-Platform_.pdf

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